Project: Worked examples
Title: Fatigue damage (Palmgren–Miner)
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Cumulative fatigue damage of a stress-range spectrum against an S–N curve of one slope through a reference point, with an optional endurance limit below which a range does no damage; the spectrum repetitions to failure.

WHAT IT CALCULATES	RELATION
Cycles to failure at each stress range, from the S–N curve (unbounded at or below the endurance limit S_e)	$N_i = N_{ref} \cdot (S_{ref}/S_i)^m$
Damage done by each block of cycles	$d_i = n_i/N_i$
Total damage, and how many repetitions of the spectrum the part survives	$D = \Sigma\, d_i$; repetitions = 1/D
Acceptance by the Palmgren–Miner rule	$D \le 1$

Basis: Fatigue resistance (S–N curves by detail category)[1](§8); Cumulative fatigue damage[2](§6-15).

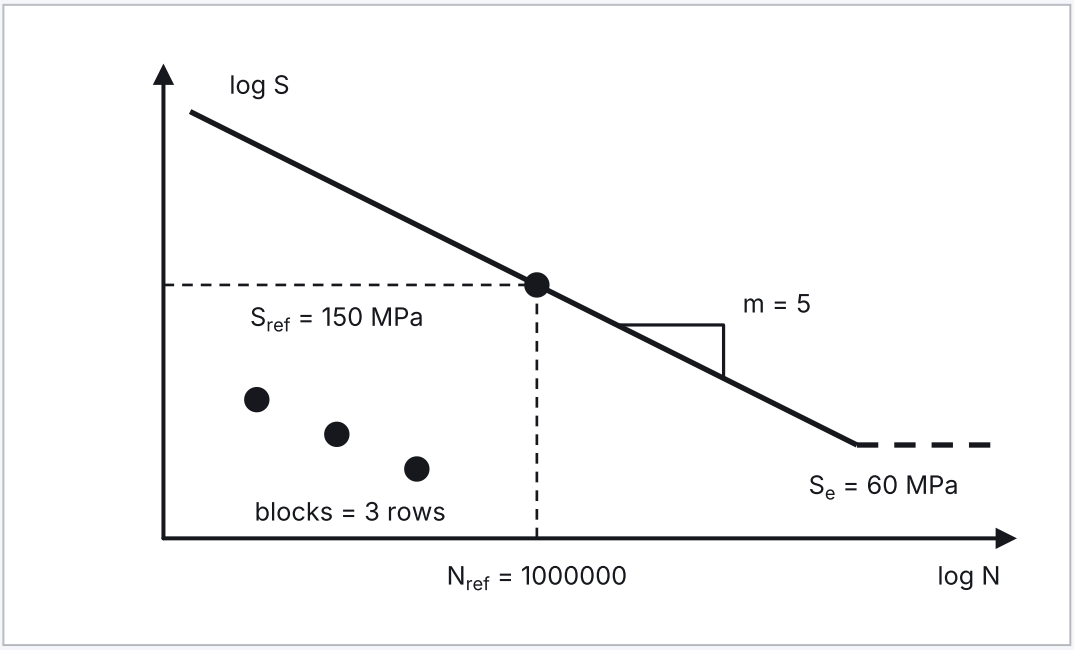


Figure 1 — A single-slope S–N line on log axes through its reference point, the endurance limit as a cut-off, and the stress-range blocks as listed (three drawn, representatively). Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
S_{ref}	Reference stress range (detail category or material fatigue strength)	150	MPa	spec: SPEC-MECH-03 design criteria	1 … 2000 MPa

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
N_{ref}	Cycles at the reference stress range	1000000		spec: SPEC-MECH-03 design criteria	10000 … 1000000000
m	S–N slope	5		spec: SPEC-MECH-03 design criteria	2 … 15
S_e	Endurance limit (ranges at or below it do no damage)	60	MPa	spec: SPEC-MECH-03 design criteria	0.1 … 2000 MPa
blocks	Stress-range blocks	3 row(s)		spec: LOAD-SCHEDULE-01	any

TABLE ROW	VALUES
blocks[1]	S = 120 MPa; n = 2000000
blocks[2]	S = 90 MPa; n = 10000000
blocks[3]	S = 50 MPa; n = 10000000

3. Assumptions

A1. Damage accumulates linearly and independently of the order of the blocks. (Palmgren–Miner rule.)
A2. One S–N slope over the whole range; no second slope below a knee. (Simplified curve; a two-slope curve is entered as two sheets or a lower slope.)

4. Working

[5] $N_1 = N_{ref} \cdot (S_{ref} / S_1)^m$ [1](§8)
= 1000000 × (150 MPa / 120 MPa)^5
= 3051800
Life at block 1

[6] $d_1 = n_1 / N_1$
= 2000000 / 3051800
= 0.65536
Damage of block 1

[7] $N_2 = N_{ref} \cdot (S_{ref} / S_2)^m$ [1](§8)
= 1000000 × (150 MPa / 90 MPa)^5
= 12860000
Life at block 2

[8] $d_2 = n_2 / N_2$
= 10000000 / 12860000
= 0.7776
Damage of block 2

[9] $N_3 = S_3 \le S_e$
= 50 MPa ≤ 60 MPa
= "unbounded"
Life at block 3: below the endurance limit

[10] $d_3 = n_3 / N_3$
= 10000000 / ∞
= 0
Damage of block 3

[11] $D = \Sigma\ d_i$ [2](§6-15)

$= 0.65536 + 0.7776 + 0$

$= 1.433$

Total damage (Palmgren–Miner)

[12] $R_f = 1 / D$

$= 1 / 1.433$

$= 0.69786$

Spectrum repetitions to failure

5. Results

SYMBOL	VALUE	DESCRIPTION
D	1.433	Total damage
R_f	0.69786	Repetitions to failure
N_1	3051800	Life at block 1
d_1	0.65536	Damage of block 1
N_2	12860000	Life at block 2
d_2	0.7776	Damage of block 2
N_3	"unbounded"	Life at block 3
d_3	0	Damage of block 3

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
FAIL	Cumulative damage D ≤ 1	D = 1.433	D ≤ 1	1.43	[2](§6-15)

7. Applicability

No warnings.

8. Verification

Template mech.fatigue-miner@1.0.1 Content hash 82e9e2cff5d85046 Engine calc-engine/1.0.0
Known-answer cases passed 2/2 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. British Standards Institution. Eurocode 3: Design of Steel Structures—Part 1-9: Fatigue. BS EN 1993-1-9. BSI.
2. Budynas RG, Nisbett JK. Shigley's Mechanical Engineering Design. 11th ed. McGraw-Hill Education; 2020.